

413/613 Fall 2026 - Applied Statistics & Data Science I

Prof. Suryanarayana Maddu

August 2026

1 Probability, Uncertainty, and Counting

1.1 Why Probability and Statistics?

Many mathematical models describe situations in which the outcome is known once the inputs are specified. These are deterministic functions, whose outcome is fixed given an input. However, in many real-world problems, we must reason in the presence of **uncertainty** arising from measurement errors, insufficient samples, and intrinsic randomness of the underlying system. Probability and statistics provide a mathematical framework for reasoning about **uncertainty** and learning from **data**. Although closely related, they address two complementary questions.

In **probability**, we assume that a probabilistic model is known and ask what kinds of observations it is likely to generate:

$$\boxed{\text{Model} \longrightarrow \text{Data}}$$

On the other hand, in **statistics**, we observe data and try to infer properties of the unknown process or distribution that generated it:

$$\boxed{\text{Data} \longrightarrow \text{Model}}$$

Given the ubiquitous nature of randomness and uncertainty associated with most worldly events, **probability** and **statistics** are used for many applications. A non-exhaustive list of application domains include genetics and other biological sciences, physics, fluid dynamics, quantum mechanics, finance and economics, political and social sciences, and many more.

A central challenge is that our intuition about uncertainty is often very unreliable, as most results in statistics are very counter-intuitive. It contains many examples where an intuitively plausible answer is incorrect. We therefore want a precise mathematical framework for questions of the form

What are all the possible outcomes, and how likely is a particular collection of outcomes ?

This leads naturally to the concepts of **sample spaces**, **events**, and **probabilities**.

1.2 Sample Spaces and Events

An **experiment** is any procedure whose outcome is not known beforehand. The word “experiment” is interpreted broadly: it could mean tossing a coin, rolling a dice, measuring a quantity of interest in a lab, observing a financial return, or recording the outcome of any uncertain process.

The **sample space**, denoted by S , is the set of all possible outcomes of the experiment:

$$S = \{\text{all possible outcomes}\}.$$

An **event** A is a subset of the sample space,

$$A \subseteq S.$$

For example, if we toss a coin twice,

$$S = \{HH, HT, TH, TT\}.$$

The event that both tosses are tails is

$$A = \{TT\}.$$

1.3 A Lazy Definition of Probability

Suppose that the sample space S is **finite** and that all outcomes are **equally likely**. Then the probability of an event A is

$$P(A) = \frac{|A|}{|S|} = \frac{\text{number of outcomes } \in A}{\text{total number of outcomes}}.$$

For example, for two fair coin tosses,

$$P(A = TT) = \frac{1}{4}.$$

IMPORTANT: There are two assumptions underlying the naive definition of probability:

1. The sample space is **finite**.
2. All outcomes in the sample space are **equally likely**.

Thus, whenever we use the lazy definition, we must first justify why the outcomes can be regarded as equally likely. Once this assumption is justified, estimating probabilities of events becomes a **counting problem**. The next question is therefore:

How do we efficiently count the number of possible outcomes?

1.4 Counting rule

Suppose an experiment consists of r successive choices. There are n_1 possible outcomes for the first choice, and for *each* such outcome there are n_2 possible outcomes for the second choice, and so on. If there are n_r possible outcomes for the r th choice, then the total number of possible outcomes is

$$n_1 n_2 \cdots n_r.$$

A useful way to visualize the multiplication rule is through a **tree diagram**: each successive choice creates another collection of branches.

Example. Suppose a furniture shop offer chairs with six different colors and three different materials. The number of possible color-material combination is

$$6 \times 3 = 18.$$

The number of possibilities can grow very quickly. For example, if we make 10 successive binary choices, then the number of possible outcomes are 2^{10} .

Thus, even simple experiments can have sample spaces that are far too large to enumerate explicitly. Counting rules provide compact mathematical expressions that allow us to determine their size without listing every outcome.

2 Counting, Story Proofs, and Probability Axioms

2.1 Some General Principles for Counting

Counting problems can be deceptively subtle. Before doing any algebra, it is useful to keep a few principles in mind.

1. Check simple and extreme cases. Suppose we derive a formula depending on parameters such as n and k . Before trusting it, check cases such as

$$k = 0, \quad k = 1, \quad n = 1,$$

or the simplest nontrivial example. For example,

$$\binom{n}{0} = 1,$$

because there is exactly one way to choose no objects from n objects. *Such trivial cases are often enough to reveal an incorrect counting argument.*

2. Label objects when appropriate. If a problem involves n people or objects, it is often useful to imagine that they are labeled

$$1, 2, \dots, n.$$

Even if two physical objects look identical, they may still correspond to distinct outcomes of an experiment.

3. Be careful about overcounting. For example, suppose 10 people are divided into a team of 4 and a team of 6. Choosing the team of 4 completely determines the team of 6, so

$$\binom{10}{4} = \binom{10}{6}$$

is the number of possible divisions. In contrast, suppose 10 people are divided into *two unlabeled teams* of 5. Then

$$\binom{10}{5}$$

counts every division twice: choosing the first group of five or choosing its complement gives the same pair of teams. Hence,

$$\boxed{\frac{1}{2} \binom{10}{5}}$$

is the correct count. Thus, “does order matter?” is often really a question about whether two descriptions correspond to genuinely different outcomes.

3 Binomial and Multinomial Counting

Many probability calculations begin with a simple question:

How many different outcomes are possible?

Suppose we choose r objects from n distinct objects, **without replacement**, and the **order does not matter**. The number of possible selections is the **binomial coefficient**

$$\boxed{\binom{n}{r} = \frac{n!}{r!(n-r)!}}.$$

For example, the number of possible five-card poker hands is

$$\boxed{\binom{52}{5}}.$$

3.1 Connection to Pascal’s Triangle

The same numbers appear as coefficients in the **binomial theorem**:

$$\boxed{(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}}.$$

For example,

$$(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.$$

The coefficients can be read directly from **Pascal’s triangle**:

$$\begin{array}{ccccccccc}
& & & & 1 & & & & \\
& & & & 1 & & 1 & & \\
& & & 1 & & 2 & & 1 & \\
& & 1 & & 3 & & 3 & & 1 \\
& 1 & & 4 & & 6 & & 4 & & 1 \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array} \quad \leftarrow \binom{4}{0}, \binom{4}{1}, \dots, \binom{4}{4}$$

Each entry is obtained by adding the two entries above it:

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}}.$$

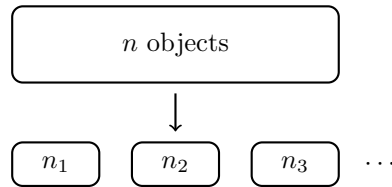
3.2 From Two Groups to Many Groups

The binomial coefficient divides n objects into **two groups**:

$$\underbrace{r}_{\text{selected}} + \underbrace{n-r}_{\text{remaining}}.$$

What if instead we divide the objects into m groups of sizes

$$n_1, n_2, \dots, n_m, \quad n_1 + \dots + n_m = n?$$



The number of possible divisions is the **multinomial coefficient**

$$\boxed{\binom{n}{n_1, n_2, \dots, n_m} = \frac{n!}{n_1! n_2! \dots n_m!}}.$$

Why? Choose each group sequentially:

$$\binom{n}{n_1} \binom{n-n_1}{n_2} \binom{n-n_1-n_2}{n_3} \dots$$

Writing out the factorials produces cancellations:

$$\frac{n!}{n_1!(n-n_1)!} \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \dots = \boxed{\frac{n!}{n_1! \dots n_m!}}.$$

The central counting principle is

$$\boxed{\begin{array}{ccc} \text{Choose } r \text{ objects from } n & \longrightarrow & \text{Divide } n \text{ objects into several groups} \\ \binom{n}{r} & & \binom{n}{n_1, \dots, n_m}. \end{array}}$$

Binomial and multinomial coefficients count possibilities; when combined with probabilities for the individual outcomes, they lead to the binomial and multinomial probability distributions.

3.3 Story Proofs: Counting the Same Thing Two Ways

Many combinatorial identities become much clearer if we interpret both sides as counting the same collection of objects. This is sometimes called a **story proof**: rather than proving an identity by manipulating factorials, we give both sides a common counting interpretation.

Example 1: Choosing a set or its complement. We have

$$\boxed{\binom{n}{k} = \binom{n}{n-k}}.$$

To choose k people from a group of n is equivalent to choosing the $n - k$ people who are *not* selected. Both sides therefore count the same thing.

Example 2: Choosing a committee and its president. Suppose we want to choose a committee of k people from n people and designate one committee member as president.

One approach is:

$$\underbrace{\binom{n}{k}}_{\text{choose committee}} \underbrace{k}_{\text{choose president}}.$$

Alternatively, first choose the president and then choose the remaining $k - 1$ committee members:

$$\underbrace{n}_{\text{choose president}} \underbrace{\binom{n-1}{k-1}}_{\text{choose remaining members}}.$$

Since both procedures count the same outcomes,

$$\boxed{k \binom{n}{k} = n \binom{n-1}{k-1}}.$$

Example 3: Vandermonde's Identity. Suppose there are two groups containing m and n people, respectively, and we want to select k people in total.

Directly, there are

$$\binom{m+n}{k}$$

ways.

Alternatively, suppose exactly j selected people come from the first group. Then $k - j$ must come from the second group. For a fixed j , there are

$$\binom{m}{j} \binom{n}{k-j}$$

possibilities.

Summing over all possible j gives

$$\boxed{\binom{m+n}{k} = \sum_{j=0}^k \binom{m}{j} \binom{n}{k-j}}.$$

This is **Vandermonde's identity**. The identity follows immediately by counting the same selection problem in two different ways.

3.4 Beyond the Naive Definition of Probability

The naive definition

$$P(A) = \frac{|A|}{|S|}$$

requires a finite sample space whose outcomes are equally likely. We now want a definition that also allows

- outcomes that are not equally likely, and
- sample spaces with infinitely many possible outcomes.

As discussed before, the **probability space** consists of a sample space S together with a probability function P that assigns to each event $A \subset S$ a number

$$P(A) \in [0, 1].$$

Thus, we may think of

$$P : \{\text{events in } S\} \longrightarrow [0, 1].$$

3.5 Axioms of Probability

The probability function $P(A)$ satisfies the following fundamental rules.

Axiom : Impossible and certain events.

$$P(\emptyset) = 0, \quad P(S) = 1.$$

The empty event is impossible, whereas some outcome in the sample space must occur.

Axiom 2: Additivity for disjoint events. If

$$A_1, A_2, A_3, \dots$$

are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

For two disjoint events A and B , this reduces to

$$P(A \cup B) = P(A) + P(B), \quad A \cap B = \emptyset.$$

4 Random Variables and Probability Distributions

A sample space Ω describes *everything that can happen*. Often, however, we do not care about the complete outcome ω . Instead, we want to extract some numerical quantity from it.

A **random variable** is a function that assigns a number to each outcome:

$$X : \Omega \longrightarrow \mathbb{R}, \quad \omega \longmapsto X(\omega).$$

Example: Number of Heads

Flip a fair coin four times. The sample space contains

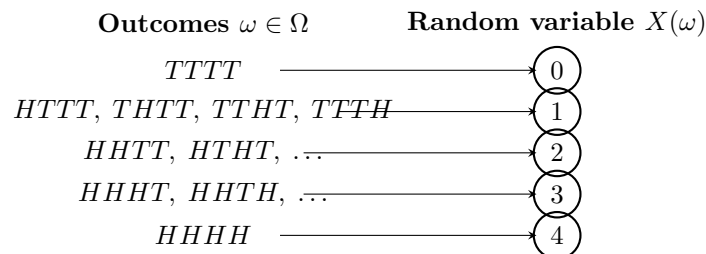
$$|\Omega| = 2^4 = 16$$

possible sequences. Define

$$X = \text{number of heads in four flips.}$$

Although Ω contains 16 outcomes, the random variable has only five possible values:

$$X \in \{0, 1, 2, 3, 4\}.$$



Many different outcomes can therefore produce the **same value** of the random variable.

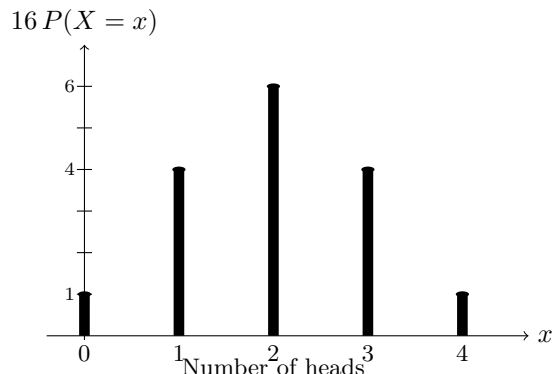
Probability Distribution

We can now ask: *how likely is each value of X ?*

For four fair coin flips,

x	0	1	2	3	4
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

This is the **probability distribution** of X .



Thus, instead of keeping track of all 16 sequences, we summarize the experiment using only the distribution of X .

For n fair coin flips,

X = number of heads

has a **binomial distribution**:

$$P(X = k) = \binom{n}{k} \left(\frac{1}{2}\right)^n, \quad k = 0, \dots, n.$$

Discrete vs. Continuous Random Variables

Random variables come in two important types:

Discrete: X takes countable values, e.g. number of heads,
Continuous: X takes values on a continuum, e.g. height.

For a discrete random variable, probabilities are obtained by **sums**, whereas for a continuous random variable with density $f_X(x)$, probabilities over intervals are obtained by **integrals**:

$$P(a < X < b) = \int_a^b f_X(x) dx.$$

A random variable converts complicated random outcomes into numbers:

$$\underbrace{\omega \in \Omega}_{\text{outcome}} \xrightarrow{X} \underbrace{x = X(\omega)}_{\text{numerical value}} \rightarrow \underbrace{P(X = x)}_{\text{distribution}}.$$

The probability distribution summarizes how likely the possible values of the random variable are.

5 Expectation and Variance: Properties and Derivations

Let X and Y be random variables and $a, c \in \mathbb{R}$ constants. Recall, for a discrete random variable,

$$\mathbb{E}[X] = \sum_x x P(X = x),$$

with the analogous definition $\mathbb{E}[X] = \int x f_X(x) dx$ for continuous random variables.

Properties of Expectation

1. Scaling by a constant.

$$\begin{aligned}\mathbb{E}[aX] &= \sum_x (ax)P(X = x) \\ &= a \sum_x xP(X = x) = \boxed{a \mathbb{E}[X]}.\end{aligned}$$

2. Adding a constant.

 Since $\sum_x P(X = x) = 1$,

$$\begin{aligned}\mathbb{E}[X + c] &= \sum_x (x + c)P(X = x) \\ &= \sum_x xP(X = x) + c \sum_x P(X = x) \\ &= \boxed{\mathbb{E}[X] + c}.\end{aligned}$$

3. Linearity of expectation.

 For *any* random variables X, Y ,

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_{x,y} (x + y)P(X = x, Y = y) \\ &= \sum_x xP(X = x) + \sum_y yP(Y = y) \\ &= \boxed{\mathbb{E}[X] + \mathbb{E}[Y]}.\end{aligned}$$

Importantly, **independence is not required**.

4. Products of independent random variables.

 If X and Y are independent, $P(X = x, Y = y) = P(X = x)P(Y = y)$, and therefore

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{x,y} xyP(X = x, Y = y) \\ &= \left(\sum_x xP(X = x) \right) \left(\sum_y yP(Y = y) \right) \\ &= \boxed{\mathbb{E}[X]\mathbb{E}[Y]}.\end{aligned}$$

Variance

Variance measures the spread of a random variable around its mean:

$$\boxed{\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]}.$$

Let $\mu = \mathbb{E}[X]$. Expanding,

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \boxed{\mathbb{E}[X^2] - \mathbb{E}[X]^2}.\end{aligned}$$

The standard deviation is

$$\boxed{\sigma_X = \sqrt{\text{Var}(X)}},$$

and has the same physical units as X .

Properties of Variance

1. Scaling.

$$\begin{aligned}\text{Var}(aX) &= \mathbb{E}[(aX - \mathbb{E}[aX])^2] \\ &= \mathbb{E}[a^2(X - \mathbb{E}[X])^2] \\ &= \boxed{a^2 \text{Var}(X)}.\end{aligned}$$

2. Adding a constant.

$$\begin{aligned}\text{Var}(X + c) &= \mathbb{E}[(X + c - \mathbb{E}[X + c])^2] \\ &= \mathbb{E}[(X + c - \mathbb{E}[X] - c)^2] \\ &= \boxed{\text{Var}(X)}.\end{aligned}$$

Adding a constant shifts the distribution but does not change its spread.

3. Variance of a sum. Define the covariance

$$\boxed{\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}.$$

Then

$$\begin{aligned}\text{Var}(X + Y) &= \mathbb{E}\left[\left((X - \mathbb{E}[X]) + (Y - \mathbb{E}[Y])\right)^2\right] \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).\end{aligned}$$

Hence

$$\boxed{\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)}.$$

If X and Y are independent,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \implies \text{Cov}(X, Y) = 0,$$

so

$$\boxed{\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)}.$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad \text{always,}$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are independent.}$$

Expectation is **linear**; variance is generally **not**.

6 Sample Means, the Law of Large Numbers, and the CLT

Suppose we observe independent and identically distributed random variables

$$X_1, X_2, \dots, X_N \stackrel{\text{iid}}{\sim} P,$$

with population mean and variance

$$\boxed{\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2}.$$

The **sample mean** is

$$\boxed{\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i}.$$

Although $\bar{X}_N$ is computed from the data, it is itself a **random variable**: a different set of N random samples will produce a different value of $\bar{X}_N$.

Mean and Variance of the Sample Mean

By linearity of expectation,

$$\mathbb{E}[\bar{X}_N] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[X_i] = \boxed{\mu}.$$

Thus, averaging does not change the expected value. For independent observations,

$$\begin{aligned} \text{Var}(\bar{X}_N) &= \text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) \\ &= \frac{1}{N^2} \sum_{i=1}^N \text{Var}(X_i) = \boxed{\frac{\sigma^2}{N}}. \end{aligned}$$

Therefore, the standard deviation is

$$\boxed{\text{SD}(\bar{X}_N) = \frac{\sigma}{\sqrt{N}}}.$$

Averaging has two fundamental effects:

$$\boxed{\text{center stays at } \mu, \quad \text{spread shrinks as } \frac{1}{\sqrt{N}}.}$$

Thus, larger samples produce increasingly stable estimates of the population mean.

Law of Large Numbers

What happens as we collect more and more observations? Since

$$\text{Var}(\bar{X}_N) = \frac{\sigma^2}{N} \rightarrow 0,$$

the sample mean becomes increasingly concentrated around μ . The **Law of Large Numbers (LLN)** formalizes this idea:

$$\boxed{\bar{X}_N \xrightarrow{P} \mu \quad \text{as } N \rightarrow \infty.}$$

In words,

$$\boxed{\text{with enough independent observations, } \bar{X}_N \text{ approaches the population mean.}}$$

Central Limit Theorem

The LLN tells us *where* $\bar{X}_N$ goes. But what does the *distribution* of $\bar{X}_N$ look like? The **Central Limit Theorem (CLT)** states that, under standard conditions,

$$\boxed{\frac{\bar{X}_N - \mu}{\sigma/\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1).}$$

Therefore, for sufficiently large N ,

$$\boxed{\bar{X}_N \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right)}.$$

Importantly, the individual observations X_i do *not* need to be Gaussian.

The LLN and CLT answer two complementary questions:

$$\begin{array}{ll} \text{LLN: } \bar{X}_N \rightarrow \mu & \text{Where does the sample mean go?} \\ \text{CLT: } \bar{X}_N \approx \mathcal{N}(\mu, \sigma^2/N) & \text{How does it fluctuate around } \mu? \end{array}$$

The CLT does **not** say that the original data become Gaussian. It says that the properly standardized **sample mean** becomes approximately Gaussian.

7 Independence and Conditional Probability

7.1 Independence

We have used the word “independent” informally. We now give it a precise mathematical definition. Two events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

The intuition is that knowing whether B occurred provides no information about whether A occurred, and vice versa.

7.2 Independent vs. Disjoint

Independence should **not** be confused with disjointness.

If A and B are disjoint, then

$$A \cap B = \emptyset,$$

and therefore

$$P(A \cap B) = 0.$$

Knowing that A occurred tells us that B *cannot* occur.

By contrast, if A and B are independent, knowing that A occurred provides no information about whether B occurs.

Disjoint	:	A occurs $\implies B$ cannot occur,
Independent	:	A occurs $\implies$ no change in our information about B .

In fact, if A and B are disjoint and both have positive probability, then they cannot be independent.

7.3 Independence of Several Events

For three events A, B, C , **mutual independence** requires

$$P(A \cap B) = P(A)P(B),$$

$$P(A \cap C) = P(A)P(C),$$

$$P(B \cap C) = P(B)P(C),$$

and also

$$P(A \cap B \cap C) = P(A)P(B)P(C).$$

The first three conditions alone are called **pairwise independence**.

Pairwise independence does *not* necessarily imply mutual independence.

More generally, events $A_1, \dots, A_n$ are mutually independent if, for *every* subset of the events,

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k P(A_{i_j}).$$

7.4 Why Conditional Probability?

Probability describes uncertainty. But in real problems, our information changes.

We perform an experiment and observe a result. We collect new data. We learn that some event has occurred. Our probabilities should therefore change to reflect this new information.

This leads to one of the central questions of statistics:

How should we update our probabilities when we receive new information or evidence?

Conditional probability provides the mathematical framework for answering this question.

Suppose we initially have some uncertainty about an event A . We then learn that another event B has occurred. We write

$$P(A \mid B)$$

for the probability of A **given** that B occurred.

7.5 Conditional Probability

For events A and B with $P(B) > 0$, the **conditional probability of A given B** is defined by

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Conditioning on B means that we restrict our attention to the part of the sample space in which B occurred. Within this restricted sample space, the part corresponding to A is precisely

$$A \cap B.$$

The division by $P(B)$ simply **renormalizes** the probabilities so that the total probability of the new sample space B is one:

$$P(B \mid B) = \frac{P(B)}{P(B)} = 1.$$

7.6 A Frequency Interpretation

Imagine repeating an experiment many times.

First retain only those repetitions in which B occurred. Among those experiments, ask what fraction also resulted in A .

Then

$$P(A \mid B) \approx \frac{\#\{\text{trials where both } A \text{ and } B \text{ occur}\}}{\#\{\text{trials where } B \text{ occurs}\}}.$$

This is the frequency interpretation of

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Thus, conditioning amounts to

restrict to outcomes consistent with the new information + renormalize.

7.7 The Multiplication Rule

Starting from the definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$

we immediately obtain

$$P(A \cap B) = P(B)P(A | B).$$

Similarly,

$$P(A \cap B) = P(A)P(B | A).$$

Therefore,

$$P(A \cap B) = P(A)P(B | A) = P(B)P(A | B).$$

7.8 Independence Revisited

If A and B are independent, then

$$P(A \cap B) = P(A)P(B).$$

Comparing this with

$$P(A \cap B) = P(B)P(A | B),$$

we obtain, provided $P(B) > 0$,

$$P(A | B) = P(A).$$

Thus independence has a natural interpretation:

A and B are independent precisely when learning that B occurred does not change the probability of A :

$$P(A | B) = P(A).$$

7.9 The Chain Rule

The multiplication rule extends naturally to several events.

For events $A_1, \dots, A_n$,

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_n) &= P(A_1)P(A_2 | A_1) \\ &\quad \times P(A_3 | A_1 \cap A_2) \\ &\quad \times \dots \times P(A_n | A_1 \cap \dots \cap A_{n-1}). \end{aligned}$$

This is called the **chain rule**.

Importantly, the events can be ordered in any way. Sometimes choosing a different ordering makes a probability calculation much easier.

7.10 Bayes' Rule

We have

$$P(A \cap B) = P(A)P(B | A)$$

and also

$$P(A \cap B) = P(B)P(A | B).$$

Equating the two expressions gives

$$P(B)P(A | B) = P(A)P(B | A).$$

Therefore,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

This is **Bayes' rule**.

It provides a fundamental mechanism for updating probabilities in light of new evidence:

$$\underbrace{P(A | B)}_{\text{updated probability}} = \frac{\underbrace{P(B | A)}_{\text{likelihood}} \underbrace{P(A)}_{\text{initial probability}}}{\underbrace{P(B)}_{\text{probability of evidence}}}.$$

Although Bayes' rule follows from only a few lines of algebra, it is one of the central ideas underlying statistical inference.

8 Bernoulli Trials, the Binomial Distribution, and Fluctuations

8.1 From Bernoulli Trials to the Binomial Distribution

Consider N independent trials, each having two possible outcomes:

success or failure.

Let

$$P(\text{success}) = p, \quad P(\text{failure}) = q = 1 - p.$$

Such an experiment is called a **Bernoulli trial**. Suppose we perform N independent Bernoulli trials and define the random variable

H = number of successes.

Clearly, the sample space S of H is

$$H \in \{0, 1, \dots, N\}.$$

What is the probability of observing exactly h successes? Consider first one particular sequence containing h successes and $N - h$ failures. By independence, its probability is

$$p^h q^{N-h}.$$

However, we do not care about the order in which the successes occur. There are

$$\binom{N}{h}$$

ways to choose which h of the N trials are successes. Therefore,

The probability of observing exactly h successes in N independent Bernoulli trials is

$$P(H = h) = \binom{N}{h} p^h (1 - p)^{N-h}, \quad h = 0, \dots, N.$$

We write

$$H \sim \text{Binomial}(N, p).$$

The structure of the formula is worth remembering:

$$\underbrace{\binom{N}{h}}_{\text{number of possible arrangements}} \underbrace{p^h (1 - p)^{N-h}}_{\text{probability of one arrangement}}.$$

For a fair coin, $p = q = 1/2$, so

$$P(H = h) = \binom{N}{h} \frac{1}{2^N}.$$

8.2 Normalization

A probability distribution must satisfy

$$\sum_{h=0}^N P(H = h) = 1.$$

For the binomial distribution,

$$\sum_{h=0}^N \binom{N}{h} p^h q^{N-h} = (p + q)^N = 1,$$

where we have used the binomial theorem and $p + q = 1$.

Whenever we introduce a probability distribution, we should check

$$\sum_x P(X = x) = 1$$

for a discrete random variable, or

$$\int p(x) dx = 1$$

for a continuous random variable.

8.3 Random Variables and Expectation

A **random variable** assigns a numerical value to the outcome of a random experiment. In our example,

$$H = \text{number of successes}$$

is a discrete random variable. If X is discrete with probability mass function $P(X = x)$, then

$$\mathbb{E}[X] = \sum_x x P(X = x).$$

More generally, for any function f ,

$$\mathbb{E}[f(X)] = \sum_x f(x) P(X = x).$$

If X is continuous with probability density $p(x)$, sums are replaced by integrals:

$$\mathbb{E}[f(X)] = \int f(x) p(x) dx, \quad \int p(x) dx = 1.$$

For a continuous random variable, probabilities are associated with intervals, for example

$$P(a < X < b) = \int_a^b p(x) dx.$$

Notice that

$$P(X = x) = 0$$

for any specific value of x .

8.4 Generating Functions

Directly evaluating moments such as

$$\mathbb{E}[H], \quad \mathbb{E}[H^2], \quad \mathbb{E}[H^3], \dots$$

from their definitions can become tedious. Define the **probability generating function**

$$G(z) = \mathbb{E}[z^H] = \sum_{h=0}^N P(H=h)z^h.$$

For the binomial distribution,

$$\begin{aligned} G(z) &= \sum_{h=0}^N \binom{N}{h} p^h q^{N-h} z^h \\ &= \sum_{h=0}^N \binom{N}{h} (pz)^h q^{N-h} \\ &= (q + pz)^N. \end{aligned}$$

Immediately,

$$G(1) = 1,$$

which again expresses normalization.

Differentiating,

$$G'(z) = Np(q + pz)^{N-1}.$$

Since

$$G'(1) = \sum_h hP(H=h) = \mathbb{E}[H],$$

we obtain

$$\mathbb{E}[H] = Np.$$

This has a simple interpretation:

$$\text{mean number of successes} = \text{number of trials} \times \text{probability of success}.$$

For a fair coin,

$$\mathbb{E}[H] = \frac{N}{2}.$$

8.5 Fluctuations: Variance and Standard Deviation

Knowing the mean alone is not enough. We also want to quantify how strongly a random variable fluctuates around its mean.

The **variance** of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Expanding the square gives the useful identity

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

The **standard deviation** is

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

For the binomial distribution, differentiate the generating function twice:

$$G''(1) = \mathbb{E}[H(H-1)] = N(N-1)p^2.$$

Since

$$H^2 = H(H-1) + H,$$

we have

$$\mathbb{E}[H^2] = N(N-1)p^2 + Np.$$

Therefore,

$$\text{Var}(H) = Np(1-p)$$

and

$$\sigma_H = \sqrt{Np(1-p)}.$$

Thus, while the mean grows as N ,

$$\mathbb{E}[H] \sim N,$$

the typical fluctuations grow only as

$$\sigma_H \sim \sqrt{N}.$$

Consequently, the **relative size of the fluctuations** is

$$\frac{\sigma_H}{\mathbb{E}[H]} = \sqrt{\frac{1-p}{Np}} \sim \frac{1}{\sqrt{N}}.$$

For a large number of independent trials,

$$\text{relative fluctuations} \sim \frac{1}{\sqrt{N}}.$$

Hence macroscopic averages become increasingly predictable as the number of independent observations becomes large.

8.6 Physical Example: Number Fluctuations in an Ideal Gas

Consider N particles uniformly distributed in a volume V . Let v be a smaller subvolume and let

X = number of particles inside v .

For any particular particle,

$$p = P(\text{particle lies in } v) = \frac{v}{V}.$$

Therefore,

$$X \sim \text{Binomial}\left(N, \frac{v}{V}\right).$$

The mean number of particles in the subvolume is

$$\mathbb{E}[X] = N \frac{v}{V}.$$

Writing the mean number density as

$$\rho = \frac{N}{V},$$

we obtain

$$\mathbb{E}[X] = \rho v.$$

Thus, the same mathematics that describes repeated coin tosses also describes fluctuations in the number of molecules occupying a region of a gas.

The broader lesson is

Bernoulli trials $\longrightarrow$ binomial distribution $\longrightarrow$ mean and variance $\longrightarrow$ quantification of fluctuations.

9 Fluctuations, Waiting Times, and Random Walks

9.1 Moments and Fluctuations

For a random variable X , the quantities

$$\mathbb{E}[X], \quad \mathbb{E}[X^2], \quad \mathbb{E}[X^3], \dots$$

are called the **moments** of the distribution.

The first moment gives the mean, while the second moment determines the variance:

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

For

$$H \sim \text{Binomial}(N, p), \quad q = 1 - p,$$

we found

$$\mathbb{E}[H] = Np.$$

Using the probability generating function

$$G(z) = (q + pz)^N,$$

we have

$$G''(1) = \mathbb{E}[H(H-1)] = N(N-1)p^2.$$

Since

$$H^2 = H(H-1) + H,$$

it follows that

$$\mathbb{E}[H^2] = N(N-1)p^2 + Np.$$

Hence

$$\text{Var}(H) = Npq$$

and

$$\sigma_H = \sqrt{Npq}.$$

The **relative fluctuation** is therefore

$$\frac{\sigma_H}{\mathbb{E}[H]} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}}.$$

Thus,

$$\frac{\sigma_H}{\mathbb{E}[H]} \sim \frac{1}{\sqrt{N}}.$$

As the number of independent observations becomes large, the *relative* size of fluctuations becomes small:

$$\text{relative fluctuations} \sim N^{-1/2}.$$

This is one reason why macroscopic averages can be extremely stable even when the underlying microscopic system is random.

9.2 Waiting for the First Success

We now ask a different question.

Suppose independent Bernoulli trials are repeated until the first success. Let

$$T = \text{trial on which the first success occurs.}$$

If the first success occurs on trial n , then the first $n-1$ trials must be failures followed by one success:

$$\underbrace{F F \dots F}_{n-1 \text{ failures}} S.$$

Therefore,

$$P(T = n) = q^{n-1}p, \quad n = 1, 2, \dots$$

This is the **geometric distribution**.

Its normalization follows from the geometric series:

$$\sum_{n=1}^{\infty} pq^{n-1} = \frac{p}{1-q} = 1.$$

The generating function is

$$G(z) = \sum_{n=1}^{\infty} P(T=n)z^n = \frac{pz}{1-qz}.$$

Differentiating and evaluating at $z=1$ gives

$$\mathbb{E}[T] = \frac{1}{p}.$$

For a fair coin,

$$p = \frac{1}{2} \quad \implies \quad \mathbb{E}[T] = 2.$$

The binomial and geometric distributions answer two different questions:

Binomial:	How many successes occur in N trials?
Geometric:	How many trials until the first success?

9.3 Random Walks

Coin tossing also provides one of the simplest models of stochastic motion.

Consider a particle starting at

$$X_0 = 0.$$

At every time step it moves

$$X_{n+1} = \begin{cases} X_n + 1, & \text{with probability } p, \\ X_n - 1, & \text{with probability } q = 1 - p. \end{cases}$$

This is a **one-dimensional random walk**.

Suppose that after n steps the particle has taken R steps to the right and L steps to the left. Then

$$R + L = n, \quad R - L = j,$$

if its final position is j .

Solving,

$$R = \frac{n+j}{2}, \quad L = \frac{n-j}{2}.$$

Therefore,

$$P(X_n = j) = \binom{n}{(n+j)/2} p^{(n+j)/2} q^{(n-j)/2},$$

provided

$$|j| \leq n \quad \text{and} \quad n+j \text{ is even.}$$

Otherwise,

$$P(X_n = j) = 0.$$

Thus the position of a random walker is directly related to the binomial distribution.

9.4 Mean and Fluctuations of a Random Walk

Let the individual steps be

$$Y_i = \begin{cases} +1, & \text{with probability } p, \\ -1, & \text{with probability } q, \end{cases}$$

so that

$$X_n = \sum_{i=1}^n Y_i.$$

For one step,

$$\mathbb{E}[Y_i] = p - q = 2p - 1,$$

and therefore

$$\boxed{\mathbb{E}[X_n] = n(p - q).}$$

For a fair random walk, $p = q = 1/2$, so

$$\boxed{\mathbb{E}[X_n] = 0.}$$

But zero mean position does *not* mean that the walker remains near the origin. For the fair walk,

$$\boxed{\mathbb{E}[X_n^2] = n,}$$

and hence the root-mean-square displacement is

$$\boxed{X_{\text{rms}} = \sqrt{\mathbb{E}[X_n^2]} = \sqrt{n}.}$$

Thus a typical distance traveled grows as

$$\boxed{|X_n| \sim \sqrt{n}.}$$

9.5 Random Walks and Diffusion

This square-root scaling has an important physical consequence.

For deterministic motion at constant velocity,

$$x(t) \sim t.$$

For a random walk,

$$\boxed{x_{\text{typical}}(t) \sim \sqrt{t}.}$$

Equivalently,

$$\boxed{t \sim x^2.}$$

Thus, to travel twice as far by diffusion requires approximately

$$2^2 = 4$$

times as long.

The simple random walk provides a microscopic model of diffusion:

$$\boxed{\text{independent random steps} \implies \langle X_t^2 \rangle \propto t \implies |X_t|_{\text{typical}} \propto \sqrt{t}.}$$

Randomness at the microscopic scale can therefore produce a precise and predictable law at the macroscopic scale.

10 Sample Spaces, Events, and Probability

Probability measures the likelihood that an **event** occurs. To formalize this idea, we first describe everything that *could* happen in a random experiment.

Sample Space and Outcomes

The **sample space** Ω is the set of all possible outcomes:

$$\Omega = \{\text{all possible outcomes of the experiment}\}.$$

A particular outcome is denoted by

$$\omega \in \Omega.$$

For three coin flips,

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad |\Omega| = 8.$$

An **event** is a subset of the sample space,

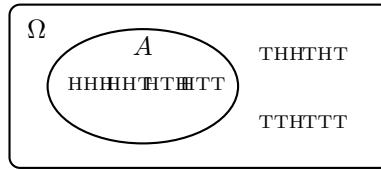
$$A \subseteq \Omega.$$

For example, if

$$A = \{\text{first flip is H}\},$$

then

$$A = \{HHH, HHT, HTH, HTT\}.$$



For equally likely outcomes,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Thus, in this example,

$$P(A) = \frac{4}{8} = \frac{1}{2}.$$

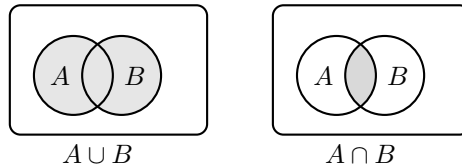
Combining Events

Let

$$A = \{\text{first flip is H}\}, \quad B = \{\text{second flip is T}\}.$$

Three fundamental set operations correspond directly to logical statements:

$$\begin{array}{ll} A \cup B : & A \text{ OR } B, \\ A \cap B : & A \text{ AND } B, \\ A^c : & \text{NOT } A. \end{array}$$



The union is more inclusive, while the intersection is more restrictive:

$$A \cap B = \{HTH, HTT\}, \quad A \cup B = \{HHH, HHT, HTH, HTT, TTH, TTT\}.$$

Hence

$$P(A \cap B) = \frac{2}{8} = \frac{1}{4}, \quad P(A \cup B) = \frac{6}{8} = \frac{3}{4}.$$

Probability as a Measure

Counting outcomes works when all outcomes are equally likely. More generally, a probability assigns a number to each event:

$$P : \{\text{events in } \Omega\} \longrightarrow [0, 1].$$

This allows, for example, biased coins where different outcomes need not have equal probability. These simple axioms imply many familiar rules:

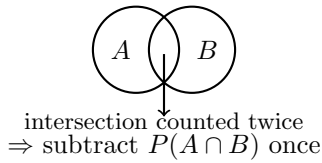
$$P(A^c) = 1 - P(A), \quad P(\emptyset) = 0,$$

and, if $A \subseteq B$,

$$P(A) \leq P(B).$$

For arbitrary events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



The basic mathematical structure of probability is

$$\text{random experiment} \longrightarrow \Omega \longrightarrow A \subseteq \Omega \longrightarrow P(A).$$

The sample space describes *what can happen*, an event specifies *what we care about*, and the probability measure quantifies *how likely that event is*.